

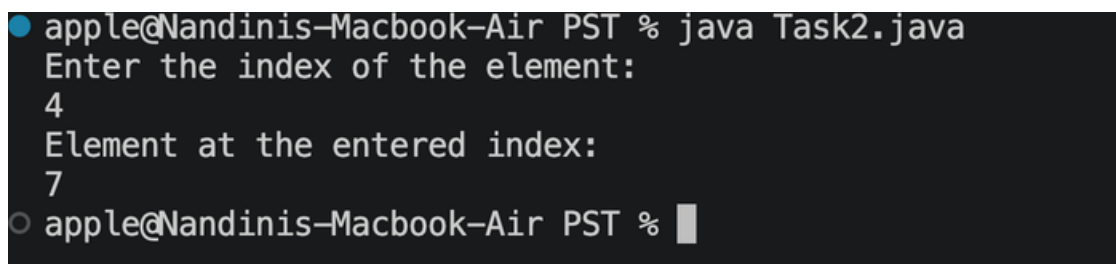
PROBLEM SOLVING AND TESTING

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WEEK 1

TASK 1:

```
import java.util.Scanner;
public class Task1 {
    public static void main(String[] args) {
        int arr[]=new int[] {1,5,3,6,7,3,4};
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the index of the element:");
        int n=s.nextInt();
        System.out.println("Element at the entered index:");
        System.out.println(arr[n]);
    }
}
```

A terminal window with a dark background. The prompt is 'apple@Nandinis-Macbook-Air PST %'. The command 'java Task2.java' has been executed. The program prompts 'Enter the index of the element:' and the user has entered '4'. The program then prompts 'Element at the entered index:' and the user has entered '7'. The prompt is now 'apple@Nandinis-Macbook-Air PST %' with a cursor.

```
● apple@Nandinis-Macbook-Air PST % java Task2.java
Enter the index of the element:
4
Element at the entered index:
7
○ apple@Nandinis-Macbook-Air PST %
```

TASK2:

```
import java.util.Scanner;

public class Task2 {
    public static int binarySearch(int arr[],int n){
        int left=0;
        int right=arr.length-1;
        while(left<=right){
            int mid=(left+right)/2;
            if(arr[mid]==n){
                return mid;
            }
            else if(arr[mid]>n){
                right=mid-1;
            }
            else
                left=mid+1;
        }
        return -1;
    }

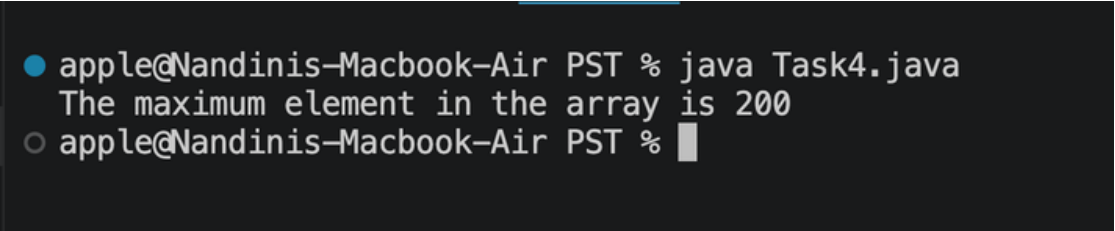
    public static void main(String[] args) {
        Scanner s=new Scanner(System.in);
        int arr[]=new int[] {23,44,56,78,79,83,89,90};
        System.out.println("Enter the element to search:");
        int n=s.nextInt();
        int result=binarySearch(arr,n);
        if(result!=-1){
            System.out.println("The element is present at the index "+result);
        }
        else{
            System.out.println("The element is not present in the array");
        }
    }
}
```

```
● apple@Nandinis-Macbook-Air PST % javac Task3.java
● apple@Nandinis-Macbook-Air PST % java Task3.java
Enter the element to search:
5
The element is not present in the array
● apple@Nandinis-Macbook-Air PST % javac Task3.java
● apple@Nandinis-Macbook-Air PST % java Task3.java
Enter the element to search:
78
The element is present at the index 3
```

TASK3:

Import java.util.Scanner;

```
public class Task3 {  
    public static void main(String[] args) {  
        Scanner s=new Scanner(System.in);  
        int arr[]=new int[]{7,67,45,34,90,178,200,44,78};  
        int max=0;  
        for(int i=0;i<arr.length;i++){  
            if(arr[i]>max){  
                max=arr[i];  
            }  
        }  
        System.out.println("The maximum element in the array is "+max);  
    }  
}
```



```
● apple@Nandinis-Macbook-Air PST % java Task4.java  
  The maximum element in the array is 200  
○ apple@Nandinis-Macbook-Air PST %
```

TASK4:

```
import java.util.*;
public class Task4{
    public static void main(String[] args) {
        Scanner s=new Scanner(System.in);
        int arr[]=new int []{4,5,3,89,76,34,98,21};
        System.out.println("Enter the number :");
        int k=s.nextInt();
        if (k > arr.length || k<= 0) {
            throw new IllegalArgumentException("Invalid value of k");
        }

        Arrays.sort(arr);
        System.out.println("The "+k+" th smallest element is "+arr[k]);
    }
}
```

```
● apple@Nandinis-Macbook-Air PST % java Task5.java
Enter the number :
4
The 4 th smallest element is 34
● apple@Nandinis-Macbook-Air PST % java Task5.java
Enter the number :
6
The 6 th smallest element is 89
○ apple@Nandinis-Macbook-Air PST %
```

TASK 5:

```
public class Task5 {  
    public static void main(String[] args) {  
        int arr[]=new int[]{1,2,3,8,6};  
        for(int i=0;i<arr.length-1;i++){  
            for(int j=1;j<arr.length;j++){  
                System.out.println "["+arr[i]+","+arr[j]+"]");  
            }  
        }  
    }  
}
```

```
apple@Nandinis-Macbook-Air PST % javac Task6.java  
apple@Nandinis-Macbook-Air PST % java Task6.java  
[1,2]  
[1,3]  
[1,8]  
[1,6]  
[2,2]  
[2,3]  
[2,8]  
[2,6]  
[3,2]  
[3,3]  
[3,8]  
[3,6]  
[8,2]  
[8,3]  
[8,8]  
[8,6]  
apple@Nandinis-Macbook-Air PST %
```

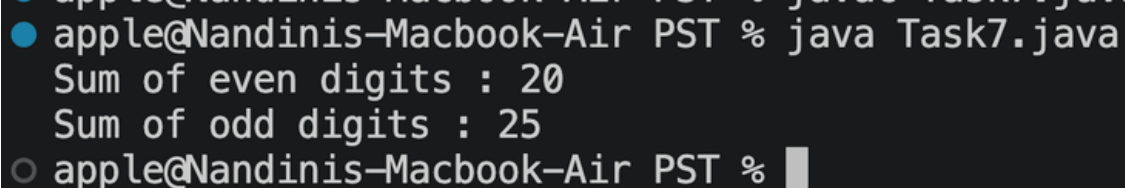
TASK 6:

```
import java.util.Arrays;
import java.util.*;
public class Task8 {
    public static void main(String[] args) {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter a number");
        int number=s.nextInt();

        int evenDigitSum = String.valueOf(number)
        .chars()
        .map(Character::getNumericValue)
        .filter(digit -> digit % 2 == 0)
        .sum();

        int oddDigitSum = String.valueOf(number)
        .chars()
        .map(Character::getNumericValue)
        .filter(digit -> digit % 2 != 0)
        .sum();

        System.out.println("Original Number: " + number);
        System.out.println("Sum of Even Digits: " + evenDigitSum);
        System.out.println("Sum of Odd Digits : " + oddDigitSum);
    }
}
```

A screenshot of a terminal window on a Mac. The prompt is 'apple@Nandinis-Macbook-Air PST %'. The user has entered 'java Task7.java'. The output shows 'Sum of even digits : 20' and 'Sum of odd digits : 25'. The prompt is now 'apple@Nandinis-Macbook-Air PST %' with a cursor.

```
apple@Nandinis-Macbook-Air PST % java Task7.java
Sum of even digits : 20
Sum of odd digits : 25
apple@Nandinis-Macbook-Air PST %
```


Task7:

```
import java.util.stream.Stream;
```

```
public class Task7 {  
    public static void main(String[] args) {  
        int n = 10;  
  
        int nthFibonacci = Stream.iterate(new int[]{0, 1}, s -> new int[]{s[1], s[0] + s[1]})  
            .limit(n)  
            .map(s -> s[0])  
            .reduce((first, second) -> second)  
            .orElse(0);  
  
        System.out.println("The " + n + "th Fibonacci number is: " + nthFibonacci);  
    }  
}
```

```
● apple@Nandinis-Macbook-Air PST % java Task8.java  
6  
The 6th fibonacci number is : 5  
● apple@Nandinis-Macbook-Air PST % java Task8.java  
7  
The 7th fibonacci number is : 8  
○ apple@Nandinis-Macbook-Air PST % █
```

TASK8:

```
public class Task8 {
    public static boolean isPalindrome(int num) {
        if (num < 0) {
            return false;
        }
        int originalNum = num;
        int reversedNum = 0;
        while (num > 0) {
            int lastDigit = num % 10;
            reversedNum = (reversedNum * 10) + lastDigit;
            num = num / 10;
        }
        return originalNum == reversedNum;
    }
    public static void main(String[] args) {
        int testNumber = 12321;

        if (isPalindrome(testNumber)) {
            System.out.println(testNumber + " is a palindrome.");
        } else {
            System.out.println(testNumber + " is not a palindrome.");
        }
    }
}
```

```
● apple@Nandinis-Macbook-Air PST % javac Task9.java
● apple@Nandinis-Macbook-Air PST % java Task9.java
12321 is a palindrome.
○ apple@Nandinis-Macbook-Air PST %
```

TASK 9 :

```
import java.util.*;
```

```
public class Task9 {  
    public static void main(String[] args) {  
        Scanner s=new Scanner(System.in);  
        System.out.println("Enter the numbers:");  
        int num1=s.nextInt();  
        int num2=s.nextInt();  
        int sum=(num1%10)+(num2%10);  
        System.out.println("The sum of last digits of two numbers is :"+sum);  
    }  
}
```

```
● apple@Nandinis-Macbook-Air PST % java Task10.java  
Enter the numbers:  
123  
456  
The sum of last digits of two numbers is :9  
○ apple@Nandinis-Macbook-Air PST %
```